n-SCRAMBLED CANTOR SETS

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ABSTRACT. For all $n \geq 2$, we provide a homeomorphism of $2^{\mathbb{N}}$ with an n-scrambled Cantor set containing an $(n+1)$-scrambled uncountable set, but with no $(n+1)$-scrambled Cantor set.

INTRODUCTION

Given a metric space X, $f\colon X \to X$, and $n \geq 2$, a sequence $x \in X^n$ is *proximal* if $\liminf_{i\to\infty} \max_{j<k<n} d_X(f^i(x(j)), f^i(x(k))) = 0$, *separated* if $\limsup_{i\to\infty} \min_{j<k<n} d_X(f^i(x(j)), f^i(x(k))) > 0$, and *Li–Yorke* if it is proximal and separated. A set $Y \subseteq X$ is *n-scrambled* if every injective sequence in Y^n is Li–Yorke. We say that Y is *$(<\mathbb{N})$-scrambled* if it is n-scrambled for all $n \geq 2$.

In [GGM21], it was shown that 2-scrambled uncountable sets yield 2-scrambled Cantor sets in Polish dynamical systems. Here we show:

Theorem 1 (ZFC). *Suppose that $n \geq 2$. Then there is a homeomorphism $T\colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ with an n-scrambled Cantor set containing a $(<\mathbb{N})$-scrambled uncountable set, but with no $(n+1)$-scrambled Cantor set.*

Let INJ^n_X denote the set of injective sequences of elements of X of length n. Let LY_f denote the set of Li-Yorke finite sequences. A *graph* on X is an irreflexive symmetric set $G \subseteq X \times X$. We say that Y is a *G-clique* if its distinct points are G-related, and *G-independent* if its points are not G-related. Let $\mathrm{CLQ}^{\geq n}_G$ denote the set of injective finite sequences of length at least n whose images are G-cliques, and let IND^n_G denote the set of injective sequences of length n whose images are G-independent. A *homomorphism* from a relation R on X to a relation S on Y is a function $\pi\colon X \to Y$ whose compositions with sequences in R are in S. Such a function is a *homomorphism* from (R, R') to (S, S') if it is also a homomorphism from R' to S'. The *saturation* of Y under a bijection $T\colon X \to X$ is given by $[Y]_T = \bigcup_{i\in\mathbb{Z}} T^i(Y)$. A subset of a topological space is F_σ if it is a countable union of closed sets. The primary new technical result underlying our proof of Theorem 1 is:

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Theorem 2 ($\mathtt{ZF}+\mathtt{DC}$)**.** *Suppose that G is an F_σ graph on $2^{\mathbb{N}}$ and $n \geq 2$. Then there exist a homeomorphism $T\colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ and a continuous injective homomorphism $\pi\colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ from $(\mathrm{CLQ}_G^{\geq n+1} \cup \mathrm{INJ}_{2^{\mathbb{N}}}^n, \mathrm{IND}_G^{n+1})$ to $(\mathrm{LY}_T, {\sim}\mathrm{LY}_T)$ for which $|{\sim}[\pi(2^{\mathbb{N}})]_T| = 1$.*

In §1, we establish Theorem 2. In §2, we obtain Theorem 1 and a related independence phenomenon as corollaries.

1. Embedding hypergraphs

Define $S\colon 2^{\mathbb{Z}} \to 2^{\mathbb{Z}}$ by $S(c)(i) = c(i+1)$ for all $i \in \mathbb{Z}$ and $c \in 2^{\mathbb{Z}}$. Let SEP_f denote the set of separated finite sequences. Theorem 2 follows from the usual topological characterization of $2^{\mathbb{N}}$ and:

Theorem 1.1 ($\mathtt{ZF}+\mathtt{DC}$)**.** *Suppose that G is an F_σ graph on $2^{\mathbb{N}}$ and $n \geq 2$. Then there exist an S-invariant perfect set $F \subseteq 2^{\mathbb{Z}}$, whose finite sequences are proximal, and a continuous injective homomorphism $\pi\colon 2^{\mathbb{N}} \to F$ from $(\mathrm{CLQ}_G^{\geq n+1} \cup \mathrm{INJ}_{2^{\mathbb{N}}}^n, \mathrm{IND}_G^{n+1})$ to $(\mathrm{SEP}_S, {\sim}\mathrm{SEP}_S)$ for which $0^{\mathbb{Z}}$ is the unique element of $F \setminus [\pi(2^{\mathbb{N}})]_S$.*

Proof. For all $s \in 2^{<\mathbb{N}}$, set $\mathcal{N}_s = \{c \in 2^{\mathbb{N}} \mid \forall i < |s| \ s(i) = c(i)\}$. For all $S \subseteq 2^{<\mathbb{N}}$, set $\mathcal{N}_S = \bigcup_{s \in S} \mathcal{N}_s$. For all $\ell \in \mathbb{N}$, $n' \geq n$, and $p\colon 2^\ell \to n'$, set $\mathcal{N}_p = \prod_{i<n'} \mathcal{N}_{p^{-1}(\{i\})}$. We say that p is a *near bijection* if it induces a bijection of $2^\ell \setminus p^{-1}(0)$ with $n' \setminus \{0\}$ but $\{0^\ell\} \subsetneq p^{-1}(\{0\})$. For all $0 < k < n'$, let $p^{-1}(k)$ denote the unique element of $p^{-1}(\{k\})$.

Fix an increasing sequence $(G_m)_{m\in\mathbb{N}}$ of closed graphs $2^{\mathbb{N}}$ whose union is G. Define $G_{\ell,m} = \{(c,c') \in 2^{\mathbb{N}} \times 2^{\mathbb{N}} \mid G_m \cap (\mathcal{N}_{c \restriction \ell} \times \mathcal{N}_{c' \restriction \ell}) \neq \emptyset\}$. Let P_ℓ denote the set of all near bijections $p\colon 2^\ell \to n$. Let $P_{\ell,m,n'}$ denote the set of near bijections $p\colon 2^\ell \to n'$ such that $\mathcal{N}_{p^{-1}(j)} \times \mathcal{N}_{p^{-1}(k)} \subseteq G_{\ell,m}$ for all $0 < j < k < n'$. Fix $I \subseteq \mathbb{N} \setminus \{0\}$ and a bijection $\phi\colon I \to \mathbb{N} \cup \bigcup_{\ell\in\mathbb{N}} (\{\ell\} \times P_\ell) \cup \bigcup_{\ell,m\in\mathbb{N},n'>n} (\{\ell\} \times \{m\} \times \{n'\} \times P_{\ell,m,n'})$ such that $I \cap [i, 2i + w(i)] = \{i\}$ for all $i \in I$, where $w(i) = 0$ if $\phi(i) \in \mathbb{N}$, $w(i) = n-2$ if $\phi(i) \in \{\ell\} \times P_\ell$ for some $\ell \in \mathbb{N}$, and $w(i) = (m+1)(n'-2)$ if $\phi(i) \in \{\ell\} \times \{m\} \times \{n'\} \times P_{\ell,m,n'}$ for some $\ell, m \in \mathbb{N}$ and $n' > n$. Define a continuous function $\pi\colon 2^{\mathbb{N}} \to 2^{\mathbb{Z}}$ by $\pi(c)(i) = 1$ if and only if

(1) $i = 0$;
(2) $i \in I$, $\phi(i) \in \mathbb{N}$, and $c(\phi(i)) = 1$;
(3) $\exists 0 < k < n \exists \ell \in \mathbb{N} \exists p \in P_\ell$
 $((\ell, p) = \phi(i - (k-1))$ and $p(c \restriction \ell) = k)$; or
(4) $\exists n' > n \exists 0 < k < n' \exists \ell, m \in \mathbb{N} \exists p \in P_{\ell,m,n'}$
 $((\ell, m, n', p) = \phi(i - (m+1)(k-1))$ and $p(c \restriction \ell) = k)$.

Lemma 1.2. *π is injective.*

Proof. Suppose that $c, c' \in 2^{\mathbb{N}}$ are distinct. Then there exists $k \in \mathbb{N}$ with $c(k) \neq c'(k)$. Fix $i \in I$ for which $\phi(i) = k$. As $\pi(c)(i) = c(k)$ and $\pi(c')(i) = c'(k)$, it follows that $\pi(c)(i) \neq \pi(c')(i)$, so $\pi(c) \neq \pi(c')$. □

The *product metric* on $2^{\mathbb{Z}}$ is given by $\rho(c, c') = \sum_{i \in \mathbb{Z}} |c(i) - c'(i)|/2^{|i|}$ for all $c, c' \in 2^{\mathbb{Z}}$. Define $N_i = [i, i + w(i)]$ for all $i \in I$.

Lemma 1.3. *π is a homomorphism from $\mathrm{CLQ}_G^{\geq n+1} \cup \mathrm{INJ}_{2^{\mathbb{N}}}^n$ to SEP_S.*

Proof. Suppose that $c \in (2^{\mathbb{N}})^{n'}$ is in $\mathrm{CLQ}_G^{\geq n+1} \cup \mathrm{INJ}_{2^{\mathbb{N}}}^n$. By permuting the coordinates of c, we can assume that $c(0) = 0^{\mathbb{N}}$ if $0^{\mathbb{N}} \in c(n')$. Suppose that $\ell \in \mathbb{N}$ is sufficiently large that $\mathrm{proj}_{2^\ell} \circ c$ is injective, $c(k) \restriction \ell \neq 0^\ell$ for all $0 < k < n'$, and $n' < 2^\ell$.

If $n' = n$, then let p be the near bijection extending $(\mathrm{proj}_{2^\ell} \circ c)^{-1}$ in P_ℓ. Fix $i \in I$ such that $\phi(i) = (\ell, p)$. Then $\mathrm{proj}_{2^{N_i}} \circ \pi \circ c$ is injective, so $\min_{j<k<n} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{n-2}$.

If $n' > n$, then fix $m \in \mathbb{N}$ such that $c(j) \; G_m \; c(k)$ for all $j < k < n'$, and let p be the near bijection extending $(\mathrm{proj}_{2^\ell} \circ c)^{-1}$ in $P_{\ell,m,n'}$. Fix $i \in I$ such that $\phi(i) = (\ell, m, n', p)$. Then $\mathrm{proj}_{2^{N_i}} \circ \pi \circ c$ is injective, so $\min_{j<k<n'} \rho(S^i(\pi(c(j))), S^i(\pi(c(k)))) \geq 1/2^{(m+1)(n'-2)}$. □

Lemma 1.4. *π is a homomorphism from IND_G^{n+1} to $\sim\mathrm{SEP}_S$.*

Proof. Suppose that $c \in (2^{\mathbb{N}})^{n+1}$ and $\pi \circ c$ is separated. Fix $\epsilon > 0$ such that $I' = \{i' \in \mathbb{N} \mid \min_{j<k<n+1} \rho(S^{i'}(\pi(c(j))), S^{i'}(\pi(c(k)))) \geq \epsilon\}$ is infinite, as well as $r \in \mathbb{N}$ such that $\sum_{|i|>r} 1/2^{|i|} < \epsilon$. By throwing out finitely many elements of I', we can assume that for all $i' \in I'$, there is at most one $i \in I$ such that $[i'-r, i'+r] \cap N_i \neq \emptyset$. Our choice of r then yields a unique such $i \in I$, in which case $\mathrm{proj}_{2^{N_i}} \circ \pi \circ c$ is injective. It follows that $\phi(i)$ is of the form (ℓ, m, n', p), where $(m+1)(n-1) \leq 2r$, and there exist $j < k < n+1$ such that $c(j) \; G_{\ell,m} \; c(k)$.

By passing to an infinite subset of I', we can assume that the values of j, k, and m are independent of $i' \in I'$. Then there are unboundedly many values of ℓ corresponding to the remaining $i' \in I'$, since otherwise there are only finitely many possibilities for both n' and p, and therefore only finitely many possibilities for i, despite the fact that the distance between any two elements of I' corresponding to the same value of i is at most $2r$. Hence $c(j) \; G_m \; c(k)$, so $c(j) \; G \; c(k)$, thus $c \notin \mathrm{IND}_G^{n+1}$. □

The closure F of the S-saturation of $\pi(2^{\mathbb{N}})$ is clearly perfect.

Lemma 1.5. *The sequence $0^{\mathbb{Z}}$ is the unique element of $F \setminus [\pi(2^{\mathbb{N}})]_S$.*

Proof. If $(c_k)_{k \in \mathbb{N}}$ is a sequence of elements of $2^{\mathbb{N}}$ and $(i_k)_{k \in \mathbb{N}}$ is a strictly increasing sequence of elements of $\mathbb{N}$, then $S^{-i_k}(\pi(c_k)) \to 0^{\mathbb{Z}}$. And if

$S^{i_k}(\pi(c_k))$ converges to a sequence $c \in 2^{\mathbb{Z}} \setminus \{0^{\mathbb{Z}}\}$, then $c(i) = 1$ for a unique $i \in \mathbb{Z}$, so it is in the S-orbit of $\pi(0^{\mathbb{N}})$. □

Lemma 1.5 implies that the union of the supports of finitely many elements of F is contained in a finite union of translates of $\{0\} \cup \bigcup_{i \in I} N_i$. But such sets necessarily have arbitrarily long gaps, so every finite sequence of elements of F is proximal. □

Proposition 1.6 ($\mathtt{ZF} + \mathtt{DC}$)**.** *Suppose that $n \geq 3$, X and Y are Polish spaces, G is a Borel graph on X, $T \colon Y \to Y$ is a Borel automorphism, $B \subseteq Y$ is an uncountable Borel n-scrambled set, $\pi \colon X \to Y$ is an injective Borel homomorphism from IND^n_G to $\sim\mathrm{LY}_T$, and $[\pi(X)]_T$ is co-countable. Then G has a perfect clique.*

Proof. Fix $i \in \mathbb{Z}$ for which $B \cap (T^i \circ \pi)(X)$ is uncountable. The perfect set theorem and Galvin's Theorem (see, for example, [Kec95, Theorems 13.6 and 19.7]) then yield a perfect set $P \subseteq (T^i \circ \pi)^{-1}(B)$ that is either a G-clique or G-independent. But G-independence would contradict the fact that B is n-scrambled. □

2. n-Scrambled sets

We begin this section with the proof of our primary result:

Proof (of Theorem 1). By [KV12, Theorem 2.1] and a straightforward Baire category argument, there is an F_σ graph G on $2^{\mathbb{N}}$ with an uncountable clique $Y \subseteq 2^{\mathbb{N}}$ but no perfect clique. By Theorem 2, there exist a homeomorphism $T \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ and a continuous injective homomorphism $\pi \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ from $(\mathrm{CLQ}^{\geq n+1}_G \cup \mathrm{INJ}^n_{2^{\mathbb{N}}}, \mathrm{IND}^{n+1}_G)$ to $(\mathrm{LY}_T, \sim\mathrm{LY}_T)$ for which $|\sim[\pi(2^{\mathbb{N}})]_T| = 1$. Then $\pi(2^{\mathbb{N}})$ is an n-scrambled Cantor set, $\pi(Y)$ is a $(<\mathbb{N})$-scrambled uncountable set, and Proposition 1.6 rules out the existence of an $(n+1)$-scrambled Cantor set. □

Let $\mathfrak{c}$ denote the cardinality of the continuum.

Theorem 2.1 ($\mathtt{ZFC} + \neg\mathtt{CH}$)**.** *It is consistent that, for all $n \geq 2$, there is a homeomorphism $T \colon 2^{\mathbb{N}} \to 2^{\mathbb{N}}$ that has an n-scrambled Cantor set containing a $(<\mathbb{N})$-scrambled set of cardinality $\mathfrak{c}$, but does not have an $(n+1)$-scrambled Cantor set.*

Proof. By [She99, Theorem 1.13], we can assume that there is an F_σ graph G on $2^{\mathbb{N}}$ with a clique of cardinality $\mathfrak{c}$ but no perfect clique. The proof of Theorem 1 therefore yields the desired result. □

A *hypergraph* is a permutation-invariant set of injective sequences.

Theorem 2.2 ($\mathtt{ZFC} + \neg\mathtt{CH}$). *It is consistent that, for all $n \geq 2$, every Borel function on a Polish space with an n-scrambled set of cardinality $\aleph_2$ has an n-scrambled perfect set.*

Proof. By [KS03, Proposition 3.4], we can assume that every Borel n-ary hypergraph on a Polish space with a clique of cardinality $\aleph_2$ has a perfect clique. But n-ary Li–Yorkeness is a Borel hypergraph. $\square$

Remark. The proof of [KS03, Proposition 3.4] can be used to establish the analog of Theorem 2.2 for ($<\mathbb{N}$)-scrambled sets.

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